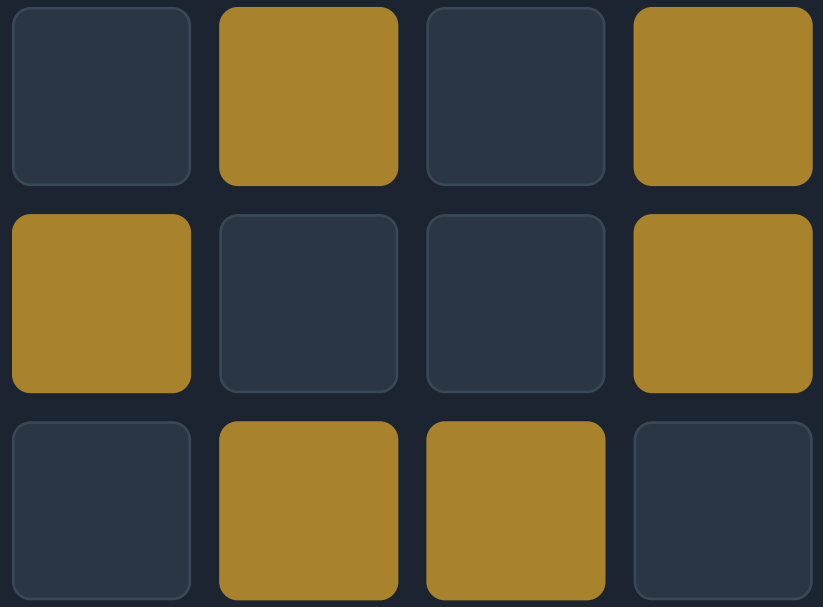
ONLINE LEARNING MANAGEMENT SYSTEM

Coursework

A full-stack MERN platform connecting Admins, Faculty, and Students around one classroom experience.

MongoDB · Express.js · React.js · Node.js · JWT · Multer



Project Presentation

INTRODUCTION

Bringing the classroom online

Coursework is an online Learning Management System built on the MERN stack. It gives students a single place to enroll in courses, study materials, and watch video lectures, while faculty create courses, upload resources, and evaluate submissions — all through secure, role-based dashboards.

3 User roles	**12** Core modules
9 Data collections	**REST** API architecture

Purpose: demonstrate real-world command of authentication, RESTful APIs, file handling, and CRUD-driven UI in a single cohesive product.

OBJECTIVES

What the platform sets out to do

01

Centralize the classroom

Replace scattered email, drive links, and spreadsheets with one platform for content, submissions, and grades.

02

Separate concerns by role

Give Admins, Faculty, and Students distinct, secure dashboards suited to what each actually needs to do.

03

Make progress visible

Let students see exactly how far they've gotten in a course, and let faculty see it too.

04

Prove the full stack

Exercise authentication, RESTful APIs, file uploads, and CRUD together in one coherent build.

SCOPE

Key modules

- Home Page
- Student Registration & Login
- Faculty Registration & Login
- Course Management
- Course Enrollment
- Study Materials
- Video Lectures
- Assignment Submission
- Quiz & Assessment (optional)
- Student Progress Tracking
- Notifications
- Contact Us

ACCESS MODEL

Three roles, one platform

Admin

- Approve & manage faculty and student accounts
- Oversee all courses across the platform
- View platform-wide enrollment & activity stats
- Resolve Contact Us inquiries

Faculty

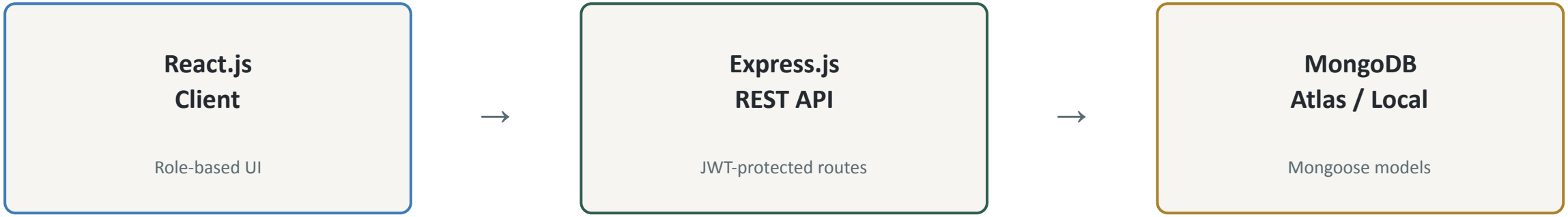
- Create and publish courses
- Upload study materials & video lectures
- Post assignments and set due dates
- Grade submissions and give feedback

Student

- Browse, search, and enroll in courses
- Access materials and stream video lectures
- Submit assignments before deadlines
- Track completion progress per course

ARCHITECTURE

Request flow, end to end



↑ file uploads (Multer) · auth tokens (JWT) · JSON payloads

Presentation	React Router pages, Context API for auth state, Axios for API calls
Application	Express controllers & routes, Multer for file storage, role middleware
Data	MongoDB collections for Users, Courses, Materials, Assignments, Progress

TECHNOLOGY STACK

MERN, end to end

React.js

FRONTEND

Component-driven UI, React Router, Context API

Node.js

RUNTIME

Server-side JavaScript execution

Express.js

BACKEND FRAMEWORK

REST routing, middleware pipeline

MongoDB

DATABASE

Mongoose schemas & document storage

JWT

AUTHENTICATION

Stateless, signed session tokens

Multer

FILE UPLOADS

Notes, videos, and assignment files

DATA MODEL

Core MongoDB collections

User

name, email, password, role, enrolledCourses

Course

title, description, category, faculty, enrolledStudents

Material

course, title, fileUrl, fileType, uploadedBy

VideoLecture

course, title, videoUrl, order, duration

Assignment

course, title, dueDate, maxMarks

Submission

assignment, student, fileUrl, marksObtained, status

Progress

student, course, completedVideos, completionPercentage

Notification

recipient, title, message, type, isRead

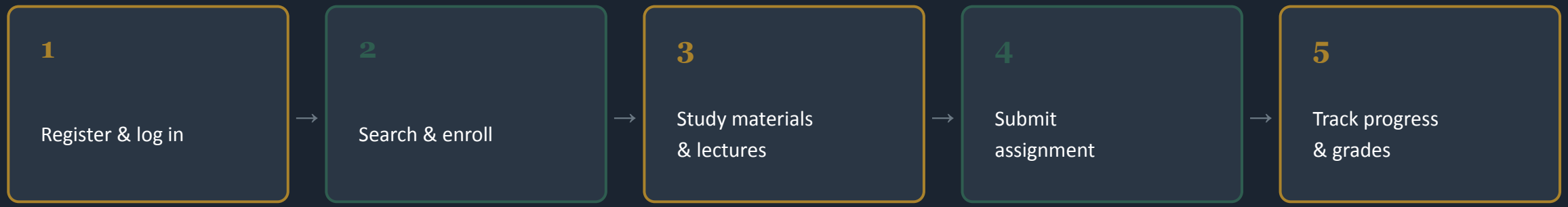
CAPABILITIES

Core features

- ✓ Secure authentication & authorization (JWT)
- ✓ Role-based access — Admin, Faculty, Student
- ✓ Full CRUD for courses & study materials
- ✓ Online assignment submission & grading
- ✓ One-click course enrollment
- ✓ Automatic progress tracking per course
- ✓ Search & filter across the course catalog
- ✓ Responsive UI across devices
- ✓ RESTful API integration throughout

WORKFLOW

A student's path through a course



Every step is backed by a corresponding REST endpoint — enrollment, material and video delivery, submission upload, and progress calculation all update MongoDB in real time.

WHY IT MATTERS

Advantages

Anywhere, anytime access

Courses, lectures, and assignments are available whenever students need them.

Paperless workflow

Assignments are submitted, graded, and returned entirely online.

Clear accountability

Role-based dashboards mean everyone sees exactly what's relevant to them.

Scalable by design

A REST API and document database make it straightforward to add courses, users, or features.

WHAT'S NEXT

Future scope

1. Live classes via integrated video conferencing
2. Auto-graded quizzes with instant feedback
3. Discussion forums per course
4. Certificates on course completion
5. Payment gateway for paid courses
6. Native mobile app (React Native)

CONCLUSION

One platform, three roles, a full stack.



Coursework demonstrates that a single, well-structured MERN application can carry real educational workflows — from enrollment to grading — securely and responsively.

Thank you